RelationasSchema

The FinTrust database has six tables. Each table stores a particular type of information. Some tables are linked so that related records can be connected without having to repeat the same information in different places.

1. Customers

Stores the personal details of customers who use the platform.

Attribute	Data type	Description
customer_id	INT	Primary key; identifies each customer
first_name	VARCHAR(50)	Customer's first name
last_name	VARCHAR(50)	Customer's last name
email	VARCHAR(100)	Customer's email address; must be unique
phone_number	VARCHAR(20)	Customer's phone number
date_registered	DATETIME	Date the customer was registered

2. Accounts

Stores the accounts created by customers. A customer can have more than one account.

Attribute	Data type	Description
account_id	INT	Primary key; identifies each account
customer_id	INT	Foreign key linked to the customers table
account_number	VARCHAR(20)	Unique account number
account_type	VARCHAR(30)	Type of account
currency	CHAR(3)	Account currency; defaults to NGN
balance	DECIMAL(15,2)	Current account balance; defaults to 0.00

account_status	VARCHAR(20)	Account status; defaults to ACTIVE
created_at	DATETIME	Date the account was created

3. Transaction_Types

Stores the different types of transactions supported by FinTrust.

Attribute	Data type	Description
transaction_type_id	INT	Primary key
type_name	VARCHAR(30)	Name of the transaction type; must be unique

The initial transaction types are Deposit, Withdrawal and Transfer.

4. Transaction_Statuses

Stores the possible statuses of a transaction.

Attribute	Data type	Description
transaction_status_id	INT	Primary key
status_name	VARCHAR(30)	Transaction status; must be unique

The statuses used in the database are Pending, Successful and Failed.

5. Transactions

Stores the financial transactions carried out on the platform. It records where the money is coming from, where it is going, the amount involved and the status of the transaction.

Attribute	Data type	Description
transaction_id	INT	Primary key
reference	VARCHAR(50)	Unique transaction reference
source_account_id	INT	Foreign key linked to the source account; can be NULL
destination_account_id	INT	Foreign key linked to the destination account; can be NULL

transaction_type_id	INT	Foreign key linked to transaction_types
transaction_status_id	INT	Foreign key linked to transaction_statuses
amount	DECIMAL(15,2)	Amount involved in the transaction
description	VARCHAR(255)	Short description of the transaction
created_at	DATETIME	Date the transaction was recorded

The source and destination account fields are allowed to be NULL because not every transaction involves two FinTrust accounts. For example, a deposit may come from outside the platform, while a withdrawal may go to an external destination.

6. Audit_Logs

Stores records of important activities carried out on the platform. This makes it possible to check what happened, when it happened and which customer was involved, where applicable.

Attribute	Data type	Description
audit_id	INT	Primary key
customer_id	INT	Foreign key linked to customers; can be NULL
action	VARCHAR(100)	Activity carried out
entity_name	VARCHAR(50)	Name of the table or record type affected
entity_id	INT	ID of the affected record
details	TEXT	Additional information about the activity
created_at	DATETIME	Date the activity was recorded

The `entity_name` and `entity_id` fields help identify the record affected by an activity. They are not linked through a normal foreign key because the record could belong to different tables.

3. Relationships Between the Tables

The tables are connected in the following ways:

- One customer can have many accounts, but each account belongs to one customer.
- One account can appear in many transactions as the source account.

- One account can also appear in many transactions as the destination account.
- One transaction type can be used by many transactions.
- One transaction status can apply to many transactions.
- One customer can have many audit log records.

These relationships help keep the database organised and make it easier to retrieve related information when needed.